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Cardiovascular care in adult cancer survivorship post-therapy: a cross-sectional analysis of clinical practice guidelines

Joshua Ayoson^{1,2}, Nadine Schneider^{1,2}, Nina Müller^{1,2}, Brian Casanova^{1,2}, Rodi Tomar^{1,2}, Lina Kleijkers^{1,2}, Luca Siragusa^{1,2}, Sacha I Rothschild³, Katharina Gut-Fischer¹, Maria M. Wertli^{§1,2,4}, Eva Haegler-Laube^{§1,5}

§Shared last authorship, equal contribution.

¹Department of Internal Medicine, Cantonal Hospital of Baden, Baden, Switzerland.

²Clinical Trial Unit, Cantonal Hospital of Baden, Baden, Switzerland.

³Division of Oncology and Hematology, Cantonal Hospital of Baden, Baden, Switzerland.

⁴Division of General Internal Medicine, University Hospital Bern, University of Bern, Bern, Switzerland.

⁵Division of Cardiology, Cantonal Hospital of Baden, Baden, Switzerland.

Corresponding author

Joshua Ayoson

Clinical Trial Unit, Department of Internal Medicine,
Cantonal Hospital of Baden, Baden, Switzerland.

Email: Joshua.ayoson@ksb.ch

Abstract

Background/Objective: Cardiovascular disease (CVD) is a major and potentially life-threatening late effect in cancer survivors, in some populations surpassing the risk of cancer recurrence as a leading cause of mortality. The aim of this study was to compare and critically appraise existing guidelines on CVD in cancer survivorship and to synthesize evidence-informed recommendations for the management of cardiovascular health in this population.

Methods: A systematic search was conducted on 16 January 2025 across PubMed, the Cochrane Library, and professional society websites. We included guidelines from professional oncology societies addressing CVD management in adult cancer survivors, published in English between 2000 and 2024. Guideline quality was assessed using the Appraisal of Guidelines for Research and Evaluation II (AGREE II) instrument. Recommendations, levels of evidence, and strengths of recommendations were extracted and standardized into the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework. Screening, grading, and data extraction were performed independently by two reviewers.

Results: Of 524 screened references, twelve guidelines from seven professional societies met the inclusion criteria and were analyzed. Of the twelve guidelines, two were rated as high quality and ten as moderate quality. Risk stratification emerged as the foundation of follow-up care, with most guidelines considering baseline cardiovascular risk, type and cumulative dose of specific cancer therapy, and treatment-related cardiotoxicity. The ESC 2022 guideline proposed the most detailed tiered model (very high risk, early high risk, late high risk, moderate risk and low risk), while others such as ASCO 2017 and NCCN 2024 guidelines used simpler schemes based on anthracycline dose, age, or pre-existing CVD. Follow-up strategies, including surveillance, pharmacologic therapy, lifestyle counseling, and specialist referral were linked to risk category. However, substantial variation existed in the timing and frequency of cardiac imaging, use of biomarkers, and thresholds for initiating cardio-protective medication. Lifestyle modification and multidisciplinary care were consistently recommended, but detail and supporting evidence varied across guidelines. Structured implementation of cardio-oncology rehabilitation models was limited. Pregnancy-related cardiovascular surveillance and long-term implications of novel oncologic therapies were inconsistently addressed. Several recommendations

were strongly endorsed despite limited prospective outcome data, highlighting a gap between recommendation strength and evidence level.

Conclusion:

In long-term survivorship, CVD represents a major determinant of morbidity and mortality beyond cancer surveillance. In this analysis, there was a consensus across guidelines on the central role of risk stratification encompassing pre- and post-treatment cardiovascular risk, treatment-related cardiotoxicity, and cardiac dysfunction during therapy in guiding follow-up care. However, the guidelines differed across oncology, cardiology, and general medicine. This variation poses a challenge, as clinicians in one field may be unfamiliar with guidelines from another. Our analysis aimed to address this gap by highlighting the coexistence of multiple, specialty-specific guidelines for the same clinical issues, such as cardiovascular disease. As guidelines reflect the evidence available at the time of publication, many do not yet account for newer oncologic therapies or the importance of structured rehabilitation programs. Harmonized, regularly updated, and evidence-based interdisciplinary recommendations are essential for the cardiovascular management of cancer survivors.

Introduction

Cancer is a leading global health challenge, accounting for 16.8% of all deaths worldwide [1]. Over the past few decades, significant advancements in cancer treatment, including immunotherapy, targeted therapy, and chemotherapy, have markedly improved patient outcomes [2]. These developments have contributed to a notable decrease in cancer-related mortality [3] and a substantial increase in the number of cancer survivors [4]. In contemporary population-based data, the 5-year relative survival rate for all cancers is approximately 70% [5], with survival rates exceeding 90% for certain cancers, such as thyroid, prostate, and testicular cancer, as well as melanoma [6]. While these improvements are commendable, they have introduced new complexities in survivorship care [7]. Survivorship care spans the entire cancer continuum, beginning at diagnosis and continuing through treatment and beyond. While the diagnostic and treatment phases focus on planning, therapy delivery, and managing acute toxicities, the post-treatment period is especially critical. Although this stage is often marked by relief at completing therapy, survivors face persistent or late-emerging effects such as cardiovascular disease (CVD), cancer-related fatigue, cognitive changes, emotional distress, and physical limitations that can significantly impact daily life [8-9]. Addressing the long-term health needs of cancer survivors remains challenging [10]. Among these challenges, CVD has emerged as a particularly pressing concern due to the cardiotoxic effects of certain cancer therapies and shared risk factors between cancer and CVD [11]. A recent study reports that cancer survivors face a substantially higher risk of CVD, particularly heart failure, compared to individuals without cancer [12]. The risk of fatal heart disease is more than two-fold higher among survivors of several solid cancers and lymphoma compared with the general population [13]. This underscores the need for individualized cardiovascular care in survivors. Primary care physicians are uniquely positioned to deliver holistic, long-term follow-up care, including recurrence monitoring, management of comorbidities, addressing psychosocial issues, and implementing preventive health measures [14, 15]. However, many primary care physicians lack the specialized expertise and structured guidance needed to manage the complex needs of cancer survivors, particularly regarding cardiovascular health, and are frequently unaware of guideline recommendations issued by the relevant specialty societies (e.g., oncology or cardiology). Moreover, even between cardiology

and oncology, there is limited awareness of survivorship-specific recommendations, with limited cross-disciplinary awareness and integration of guidance across disciplines. Although professional organizations such as the American Society of Clinical Oncology (ASCO) [16], the National Comprehensive Cancer Network (NCCN) [17], the European Society of Cardiology (ESC) [18], and others [19–21] have developed guidelines regarding survivorship care, many recommendations are based on expert consensus or limited evidence. In pediatric and adolescent cancer survivorship, standardized cardiovascular surveillance strategies have been developed by the International Late Effects of Childhood Cancer Guideline Harmonization Group [22–23] with recent updates further refining cardiomyopathy surveillance recommendations. These guidelines provide comprehensive recommendations regarding surveillance modalities, frequency, and the management of detected abnormalities, including tailored recommendations for specific clinical scenarios such as surveillance during pregnancy and the absence of routine surveillance recommendations in low-risk populations. While these are widely recognized and routinely applied in pediatric oncology, their integration and transition into adult survivorship care remain challenging. Importantly, recommendations for long-term cardiovascular care in childhood cancer survivors must be maintained following transition to adult cardiology services. To address this knowledge gap between specialties, a systematic overview of existing guidance was considered the most appropriate study design. This approach allows for structured comparison across organizations, identification of areas of consensus and divergence, and critical appraisal of the underlying evidence. Therefore, this study aimed to summarize current recommendations for cardiovascular health from different specialties (e.g., oncology and cardiology) in cancer survivorship after treatment and assess the underlying level of evidence. By synthesizing current recommendations, this study provides a foundation for the development of more cohesive, evidence-based strategies across oncology and cardiology to improve cardiovascular care in cancer survivorship.

Methods

Study design

This cross-sectional study was conducted as part of a systematic interdisciplinary framework to summarize existing recommendations for different clinical aspects of adult cancer survivorship care. Recommendations on cancer related fatigue has been published [24]. In this study, we used the same methods and systematically searched for guidelines addressing the management of cardiovascular health in cancer survivorship care. In the systematic literature search, we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to identify all relevant guidelines [25]. Results are reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist for cross-sectional studies [26] and the PRISMA statement for systematic reviews.

Systematic search of guidelines

We systematically searched the following databases on 16 January 2025: PubMed and the Cochrane Library. We identified relevant MeSH terms and keywords such as “cancer survivors,” “cancer survivorship,” “post-cancer care,” “follow-up,” and “after therapy,” which were used in combination with “guideline,” “clinical practice guideline,” and “recommendation”. The full search is depicted in **Table S1**. In addition to the systematic search, we also searched the webpages of relevant professional societies, reviews, and additional literature as well as external sources through consultation with experts in the field.

Eligibility criteria

We included all guidelines developed or endorsed by national or international oncology or survivorship care societies addressing cancer survivorship. We included guidelines that provided recommendations for the management of cancer survivors during the long-term follow-up care published between 2000 and 2024. We chose the wide timeframe to ensure we captured the full range of potentially relevant guidelines, including earlier foundational documents that may have shaped subsequent guidance on survivorship care and CVD recommendations. Survivorship is increasingly recognized as a continuum with distinct phases-acute, extended, and long-term each associated with different needs and challenges. While the acute phase extends from diagnosis through the end of primary treatment, and the extended phase spans remission or completion of treatment up to about five years post-diagnosis, the long-

term phase begins thereafter and continues until the end of life [10]. Our study focused on the long-term survivorship phase, defined as ≥ 5 years after cancer diagnosis, as this period represents a clinically meaningful inflection point in cardiovascular risk trajectories. As demonstrated in recent studies, cardiovascular risk following cancer therapy follows a biphasic pattern: an early treatment-associated risk period followed by a relative increase in cardiovascular mortality compared with primary cancer mortality beginning approximately five years after therapy [27–28]. Most cardiovascular diseases, including atherosclerosis, develop progressively over time due to cumulative exposure to cardiotoxic therapy, aging, and traditional risk factors. Accordingly, the risk of CVD-related death varies with time from cancer diagnosis, with many survivors experiencing the highest relative cardiovascular risk at ≥ 5 years after completion of curative treatment as reported by the NCCN 2024 guideline [17]. Importantly, it frequently coincides with reduced oncology-directed follow-up and transition to primary or cardiology-led care. We therefore concentrated on recommendations addressing structured long-term surveillance, risk stratification, and prevention of late cardiovascular sequelae. Recommendations initiated within the first years after treatment were included when their purpose was the early detection of progressive, late-onset toxicity. When multiple guidelines from the same society and/or several publications were available, we included the most recent. Excluded were guidelines without recommendations for cardiovascular health, other publication types (e.g., reviews, protocols, commentaries, and editorials), guidelines and recommendations for cancer survivors of childhood cancers. Although survivors treated in childhood ultimately transition into adult cardiovascular services and require lifelong surveillance [29], their management is guided by dedicated exposure-based harmonization frameworks developed for pediatric and adolescent oncology. These frameworks reflect not only different cumulative dose thresholds and cardiotoxicity risk models, but also distinct treatment paradigms and survivorship planning strategies, including front-loaded exposure considerations and decades-long risk anticipation beginning at diagnosis. Our objective was to examine guideline heterogeneity in survivors treated during adulthood, whose cardiovascular risk trajectories, treatment decision frameworks, and surveillance integration differ structurally from those treated in childhood [30]. We also excluded guidelines published before 2000 because the underlying evidence, and thus the recommendations, may be outdated.

Study procedure

The study selection process was conducted independently by two reviewers (J.A. and N.S.), who screened all titles and abstracts from the identified records to assess eligibility. In the next step, the two reviewers independently read full texts of potentially relevant references for inclusion or exclusion. All data on recommendations were extracted independently by the two reviewers using a predefined spreadsheet covering authors, society, clinical problem, intervention, and effect of intervention, level of evidence, and strength of recommendation. If a guideline used other methods to grade the evidence and strength of recommendations, the grading was standardized according to the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework [31]. Disagreements between the assessments of the two reviewers were discussed between the reviewers and resolved by consensus. If no consensus was reached, third party arbitration was done by a third reviewer (E.H-L. or M.W.).

Quality of guidelines

In compliance with the Appraisal of Guidelines for Research and Evaluation II (AGREE II) criteria [32], two reviewers (J.A and N.S.) independently assessed the quality of the guidelines (Table S2). Given that the manual recommends up to four assessors, we discussed all discrepancies and resolved them through consensus discussions by two other authors (E.H-L. and M.W.). This assessment provided an overall evaluation as well as a detailed analysis across six domains, comprising 23 individual items: scope and purpose (3 items), stakeholder involvement (3 items), rigor of development (8 items), clarity of presentation (3 items), applicability (4 items), and editorial independence (2 items). Each item was scored on a 3-star (***) scale, with 1 star indicating "not adhered to", 2 stars indicating "moderately adhered" to and 3 indicating "fully adhered to." To summarize overall quality, we applied the following classification criteria: High quality: Guidelines receiving 3 stars in at least 5 of the 6 domains (i.e., >15 total stars). Moderate quality: Guidelines receiving an average of 2 stars across domains (i.e., 12–15 total stars), including occasional domains rated as 1 star. Low quality: Guidelines with <12 total stars or multiple domains rated as 1 star. Given our focus on CVD-specific content, providing a general recommendation for the entire guideline would have been potentially misleading and outside our study scope. Instead, we evaluated; (1) the overall methodological quality of each guideline using

AGREE II, and (2) the strength and evidence base of CVD-related recommendations.

Statistical analyses

We summarized continuous variables descriptively and categorical variables using number and percentage.

Ethical Approval

We did not obtain ethical/IRB approval for our submission, as the study utilized publicly available data that did not involve human participants.

Results

Systematic literature review

After the exclusion of 105 duplicate references, we screened 420 references (Figure 1). The title and abstract screening resulted in the exclusion of 269 references. Full-text screening of 151 references resulted in the exclusion of guidelines that addressed recurrence or progression (n=79) or late effects (n=60; e.g., cancer-related fatigue or endocrine complications) but did not provide specific recommendations for CVD. Finally, twelve guidelines from seven professional societies were included in the analysis.

Guidelines and quality assessment

Guidelines were either broad in scope (n=7, 58.3%) or focused on specific cancer types (e.g., breast, colorectal, or prostate (n=5, 41.7%; **Table 1**)). These guidelines were issued by organizations from both national and international levels [16-19, 21, 34-40]. The overall quality of the cancer survivorship guidelines varied from high (n=2, 16.7%; ACS 2017 [16] and ESC 2022 [18]) to moderate (n=10, 83.3%; **Table S2**).

Specific recommendations

Risk stratification

Risk stratification in cancer survivors should take into account baseline CV risk, the type and cumulative dose of cancer therapy (particularly anthracyclines and thoracic radiotherapy), time since treatment, and the presence or recovery of cardiac

dysfunction (ESC 2022 [18]). The use of validated CV risk estimators is recommended (NCCN 2024 [17]; ESC 2022 [18]). For men receiving androgen deprivation therapy, screening according to the United States Preventive Services Task Force guidelines is specifically recommended (ACS 2014 [35]). Baseline assessment should also identify modifiable risk factors such as hypertension and dyslipidemia (NCCN 2024 guideline [17]).

Cancer therapy-related CV toxicity is particularly common in patients treated with anthracyclines (all guidelines), HER2-targeted agents, and thoracic radiotherapy (all guidelines). The ESC 2022 guideline [18] provides a detailed, tiered model for risk stratification. Survivors are categorized as very high, early high-risk, late high-risk, moderate, or low risk based on treatment exposures and CV risk factors.

Cumulative dose cut-offs for cardiotoxic treatments are summarized in Table 2. For example, anthracycline exposure (e.g., doxorubicin) and radiotherapy involving the heart (mean heart dose [MHD] >25 Gy) place survivors in the very high-risk category (strong recommendation, moderate to high level of evidence). Hematopoietic stem cell transplantation is classified as early high-risk (consensus). The late high-risk group includes survivors with moderate treatment exposures (e.g., radiotherapy 15–25 Gy, or radiotherapy 5–15 Gy combined with anthracycline ≥ 100 mg/m²) and uncontrolled CV risk factors. For lower levels of anthracycline or radiotherapy exposure, patient-specific modifiers become increasingly relevant. The ASCO 2017 guideline, for instance, incorporates factors such as age ≥ 60 years, pre-existing CV risk factors, or borderline left ventricular ejection fraction (LVEF 50–55%) when stratifying survivors (moderate level of evidence).

Surveillance

Follow-up strategies are based on risk stratification. The ESC 2022 guideline recommends annual CV risk assessment (clinical review, blood pressure and lipid profile, HbA1c), ECG, and natriuretic peptides in all survivors who were treated with potentially cardiotoxic drugs or radiation therapy (strong for, moderate level of evidence). The ESMO 2024 guideline [39] provides a conditional recommendation for cardiac follow-up as clinically indicated in asymptomatic survivors with potentially cardiotoxic treatment. According to the ESC 2022 guideline [18], risk stratification should be repeated every five years, and cardiovascular risk factors should be managed (consensus).

Transthoracic echocardiography (TTE) should be done based on the risk stratification: Very high or early high risk: TTE at one, three, and five years after treatment, with continued follow-up every five years thereafter. Node-positive, human epidermal growth factor receptor 2 (HER2)-positive breast cancer survivors with high dose exposure or additional risk factors should undergo TTE 1 year after anthracycline therapy (NCCN 2024 [17]). In a study in node-positive HER2-positive breast cancer survivors, 98% of cardiotoxicity was identified within the first year after treatment during a median 5.2-year follow-up [41]. In late high risk, TTE should be performed five years after treatment and continue at five-year intervals. In moderate-risk, TTE should be considered every five years. However, if cardiovascular risk factors are poorly controlled, reclassification to a higher-risk group and more frequent surveillance should be considered. For patients who developed cancer therapy-related CVD during treatment, as well as those in whom cardio-protective measures were initiated during or before treatment, immediate cardiologic work-up is recommended. The ESMO 2020 guideline [40] recommends that asymptomatic patients with prior cardiotoxic therapy and normal cardiac function undergo biomarker and, if appropriate, imaging-based screening at 6–12 months, 2 years, and periodically thereafter.

Coronary artery disease (CAD) screening should be used in high-risk asymptomatic survivors with non-invasive tests such as stress echocardiography, cardiac CT, stress MRI, PET stress test, according to local protocols (ESC 2022 [18]; conditional recommendation for, low level of evidence). Screening should begin 5 years after radiotherapy to the chest that included the heart (in particular, mean heart dose > 15 Gy, **Table 2**). In patients with symptoms of cardiac dysfunction, TTE as part of the initial diagnostic workup should be performed (ASCO 2017 [16]). If TTE is unavailable or technically limited, cardiac MRI or MUGA scan may be considered, with a preference for cardiac MRI. The NCCN 2024 guideline [17] also emphasized the need for timely referral in symptoms suggestive of cardiac dysfunction such as dyspnea, orthopnea, or reduced exercise tolerance.

Two guidelines recognize the clinical uncertainty regarding surveillance in asymptomatic survivors and recommend cardiac follow-up as clinically indicated without specific time intervals (ESMO 2024 [39], ASCO/ACS 2016 [34]) beyond the first-year post-treatment. The ACS 2015 [36] and ASCO/ACS 2016 [34] guidelines provide disease-specific recommendations. In colorectal cancer survivors with obesity or known coronary artery disease who have been treated with 5-fluorouracil or

capecitabine, the ACS 2015 [36] advises monitoring CRC survivors who are obese or who have had prior coronary artery disease. The 2016 guideline [34] emphasizes lipid monitoring and general cardiovascular assessment in breast cancer survivors (low level of evidence). In survivors undergoing radiotherapy to the head and neck, carotid ultrasound is conditionally recommended every 5 years (low level of evidence).

Assessment

Guidelines highlighted the importance of a careful history and physical examination to identify symptoms of cardiac dysfunction (moderate to very low level of evidence). In survivors with symptoms or signs of cardiac dysfunction, TTE should be performed to assess for underlying structural heart disease. In cases where the TTE is normal, further diagnostic evaluation should consider non-cardiac causes, with appropriate referrals to specialties such as pulmonology or cardiology. Survivors with asymptomatic cardiac dysfunction during routine surveillance or evidence of cardiomyopathy should be referred to the cardiologist or cardio-oncologist, if available (very low to moderate level of evidence, strong recommendation for).

Lifestyle Modifications and the Emerging Concept of Cardio-Oncology Rehabilitation (CORE)

Across the reviewed guidelines [16–19, 21, 34–36], lifestyle modification is consistently recommended as part of survivorship care. Most guidelines endorse regular physical activity (typically ≥ 150 minutes of moderate-intensity exercise per week), resistance training, smoking cessation, weight control, and dietary optimization. However, these recommendations are generally presented within the framework of health promotion, quality of life and behavioral counseling rather than as structured cardiovascular rehabilitation interventions.

In parallel with these recommendations, the field of cardio-oncology has progressively conceptualized exercise as a therapeutic modality within a structured rehabilitation model. The American Heart Association [42] articulated the role of cardio-oncology rehabilitation as an adaptation of traditional cardiac rehabilitation, integrating supervised exercise training, cardiovascular risk assessment, and multidisciplinary care. More recently, the ICOS CORE working group has expanded this framework, proposing Cardio-Oncology Rehabilitation and Exercise (CORE) [43] as a multimodal, risk-stratified intervention spanning the cancer continuum. CORE emphasizes

individualized exercise prescription, cardiopulmonary evaluation, optimization of cardiovascular risk factors, and coordinated oversight between oncology and cardiology services.

Recent evidence supports this evolving framework. A 2025 meta-analysis [44] demonstrated that exercise-based interventions improve cardiorespiratory fitness and selected cardiovascular risk markers in cancer populations, and a randomized trial of digital cardio-oncology tele-rehabilitation showed significant improvements in VO_2 peak and quality of life with high adherence and no safety concerns [45]. Within the guidelines analyzed in this study, explicit operationalization of a comprehensive CORE model was limited. While some guidelines recommend referral to supervised therapy or specialist assessment in selected patients, most emphasize general physical activity targets without detailing structured, multidisciplinary rehabilitation pathways. This observation likely reflects the evolving evidence base and the relatively recent formalization of CORE as a distinct clinical framework. As the evidence supporting structured CORE programs continues to mature, future guideline iterations may further clarify the role of supervised, risk-adapted rehabilitation strategies in long-term cardiovascular care for cancer survivors.

Recommendations for the Management of Specific Conditions (Table 3)

Radiation and Coronary Artery Disease (CAD)

In asymptomatic survivors who underwent non-invasive testing with CCTA, moderate or severe CAD should result in additional ischemia testing. If significant findings such as inducible ischemia or severe left main disease are present, a multidisciplinary team discussion is strongly recommended to guide treatment decisions.

In symptomatic survivors with radiation-induced CAD being considered for coronary artery bypass grafting (CABG), the ESC 2022 guideline [18] recommends pre-operative assessment of the internal mammary arteries (LIMA/RIMA), venous access, and the sternal wound healing potential due to the impact of prior radiation on surgical feasibility and outcomes (consensus). For patients with severe left main or three-vessel disease and a high SYNTAX score (>22), percutaneous coronary intervention (PCI) may be considered if technically feasible (moderate level of evidence). For patients with a history of mediastinal chest RT, evaluation for CAD,

ischaemia, and valvular disease is recommended, even if asymptomatic starting at 5 years post-treatment and then at least every 3–5 years thereafter (ESMO 2020 guideline [40]).

Radiation Associated Valvular Heart Disease (VHD)

In cancer survivors with symptomatic aortic stenosis after mediastinal radiotherapy, transcatheter aortic valve implantation (TAVI) should be considered, given the challenges posed by radiation-induced mediastinal fibrosis or calcification (ESC 2022 [18]; moderate level of evidence). As with CAD, a multidisciplinary team (MDT) approach is advised to evaluate surgical risk and to guide individualized treatment strategies in survivors with severe VHD (consensus).

Medications in cancer therapy-related cardiac dysfunction

Heart failure treatment should align with standard HF management protocols (ESC 2022 guideline [18] refers to the 2021 ESC Guidelines). Treatment with ACE inhibitors (ACE-I), angiotensin receptor blockers (ARBs), and/or beta-blockers were recommended not only for symptomatic but also for asymptomatic cancer survivors with a left ventricular ejection fraction (LVEF) <40%. Furthermore, in cases with signs for cardiac dysfunction based on a new GLS or elevated cardiac biomarkers with LVED >50%, initiation of ACE-I/ARB and/or beta-blockers may be considered (consensus). The ESMO 2020 guideline [40] recommends indefinite continuation of ACE-Is, ARBs, and/or BBs with regular cardiology follow-up in patients with LVD or HF from trastuzumab, anthracyclines, or other anticancer therapies, even if LVEF or symptoms improve. Withdrawal should only be considered after prolonged stability, absence of risk factors, and no further anticancer therapy.

Pregnancy in cancer survivors

While the ESC 2022 Guideline [18] provides specific recommendations for high-risk female survivors, pregnancy surveillance is inconsistently addressed across other guidelines. The 2022 ESC Guidelines recommend structured cardiovascular risk stratification and surveillance in this population. In high-risk survivors, particularly those with prior CTRCD, pre-pregnancy counseling and management by a multidisciplinary pregnancy heart team are recommended (Evidence: low, Recommendation: strong for). A baseline cardiovascular evaluation including history,

physical examination, ECG, natriuretic peptides, and echocardiography is recommended in women with prior CTRCD, and should be considered in all survivors exposed to potentially cardiotoxic therapy who are contemplating pregnancy. During pregnancy, echocardiographic reassessment is recommended at 12 weeks' gestation in high-risk patients or in those without prior baseline assessment, with consideration of repeat evaluation at 20 weeks in high-risk women. Therefore, while pregnancy-related cardiac events are infrequent in survivors without prior dysfunction, systematic preconception evaluation and trimester-based imaging surveillance are advised in high-risk groups to enable early detection of ventricular dysfunction and timely multidisciplinary management.

Long-term cardiotoxic effects of novel treatment options

The ESMO 2022 guideline [37] expands the spectrum of cardiotoxic therapies beyond anthracyclines and radiotherapy to include novel agents such as immune checkpoint inhibitors (ICIs), HER2-targeted therapies, VEGF inhibitors, tyrosine kinase inhibitors (TKIs), and proteasome inhibitors. While many cardiovascular toxicities associated with these therapies occur during active treatment, increasing attention is directed toward their potential long-term sequelae in cancer survivors [46]. ICIs are predominantly associated with early immune-mediated toxicities, including myocarditis and conduction disturbances. Importantly, the ESMO 2022 guideline reports that the long-term effects of ICI treatment on cardiovascular disease are unknown. Although emerging data suggests possible acceleration of atherosclerosis and an increased incidence of acute coronary syndromes following ICI exposure [47], robust long-term prospective data are lacking. HER2-targeted therapies and proteasome inhibitors may result in persistent or incompletely reversible left ventricular dysfunction in a subset of patients, particularly in those previously exposed to anthracyclines [48-49]. VEGF inhibitors and several TKIs are associated with sustained hypertension and endothelial dysfunction, potentially contributing to late-onset vascular disease and heart failure [50]. Given these considerations, long-term cardiovascular follow-up in survivors exposed to novel cardiotoxic therapies should be risk-adapted. A structured cardiovascular evaluation at completion of therapy is advisable, with periodic reassessment in high-risk individuals, particularly those with prior cancer therapy-related cardiac dysfunction, combined cardiotoxic exposures, or persistent cardiovascular risk factors. However, standardized surveillance intervals for late

toxicity following novel therapies have not yet been established, underscoring an important gap in contemporary cardio-oncology survivorship care.

Discussion

In long-term survivorship, the importance of cancer recurrence decreases, while the impact of other diseases, particularly CVD, increases. Guidelines recommend that in all cancer survivors, the individual CVD risk should be stratified based on their cardiovascular risk profile, preexisting conditions, the cardiotoxicity of the cancer-related treatment, and evidence for cardiovascular dysfunction. In particular, the ESC 2022 guideline offers the most tiered approach to care and provides a structured framework for clinical practice.

However, our analysis showed variation in risk stratification and the intensity of follow-up care, which warrant attention in future guideline development. These discrepancies appear to arise from two principal factors. First, differences in scope and target populations contribute to heterogeneity: ESC applies a broad exposure-based risk taxonomy with structured long-term surveillance intervals, whereas NCCN provides more tumor- and scenario-specific recommendations without a unified survivorship algorithm. Second, this variation reflects differing interpretations of limited prospective evidence, particularly regarding biomarker use and imaging frequency. For example, ESC integrates natriuretic peptides and scheduled echocardiography within formal follow-up pathways, while ESMO and ASCO more often frame these tools as conditional or symptom triggered. Radiation-associated coronary screening further illustrates this divergence, with ESC recommending conditional testing at 5–10-year intervals and ESMO assigning stronger recommendations to more frequent surveillance despite similar evidentiary constraints. One explanation for these differences is the limited availability of high-quality prospective outcome data. Furthermore, current survivorship surveillance algorithms are anchored in historical cardiotoxicity paradigms and do not yet adequately account for the evolving and incompletely defined long-term cardiovascular risks associated with modern immunotherapies and targeted treatments, reflecting an important gap between therapeutic innovation and guideline adaptation.

Beyond surveillance, structured exercise is increasingly supported by interventional evidence in cancer survivorship. The CHALLENGE trial [51] demonstrated improved disease-free survival with a supervised exercise program following adjuvant therapy in colon cancer survivors, alongside sustained gains in physical activity and fitness. Although cardiovascular endpoints were not evaluated, these findings reinforce the role of exercise as a structured therapeutic component of survivorship care rather than solely lifestyle recommendation. Cardiovascular-specific evidence further supports this concept. In the OptiTrain randomized trial, high-intensity exercise during anthracycline therapy was associated with lower NT-proBNP levels at one year and a reduced likelihood of biomarker-defined subclinical cardiotoxicity [52]. Complementary mechanistic evidence suggests that exercise may mitigate anthracycline-related myocardial stress through improvements in vascular function and cardiac reserve [53].

Most guidelines predate this evolving interventional evidence and continue to frame physical activity primarily within a preventive paradigm. As randomized trials increasingly incorporate biomarker and functional cardiovascular endpoints, future guideline development may need to more explicitly define the role of supervised, risk-adapted exercise within integrated oncology–cardiology care models.

Beyond exercise, several cardiovascular management domains demonstrated a clear divergence between recommendation strength and the underlying evidence. Strong endorsements of long-term echocardiographic surveillance after anthracycline therapy are primarily based on expert consensus rather than prospective outcome trials. Radiation-associated coronary and valvular screening strategies similarly lack robust validation of optimal surveillance intervals and clinical benefit. Biomarker-based monitoring remains inconsistently supported by longitudinal outcome data, and guidance regarding continuation of heart failure pharmacotherapy in recovered CTR-CVT is largely extrapolated from non-oncology populations. Together, these areas exemplify the “evidence–recommendation gap” described in contemporary cardio-oncology literature [16, 54] and define clear priorities for prospective risk model validation and outcome-driven surveillance trials.

Overall, our findings highlight heterogeneity in cardiovascular surveillance recommendations and underscore the need to better align emerging interventional evidence with multidisciplinary guideline implementation to optimize long-term cardiovascular outcomes in adult cancer survivors.

Findings in the Light of the Literature

Current cardio-oncology and survivorship guidelines from major societies including the European Society of Cardiology (ESC, 2022 [18]), the NCCN 2024 [17], and the American Society of Clinical Oncology (ASCO, 2017 [16]) provide an essential framework for cardiovascular surveillance in cancer survivors. However, in the absence of systematic reviews or harmonized recommendations, substantial variation persists in how societies define surveillance intervals, preferred imaging modalities, and thresholds for intervention. This synthesis extends the guideline-based evidence by identifying persistent gaps in follow-up strategies and outlining where survivorship care must evolve beyond current recommendations. A major gap is the lack of consensus on surveillance frequency and duration. While most societies endorse risk-adapted imaging, their recommendations diverge considerably, particularly beyond the first year after therapy. Another limitation is the incomplete integration of modern cancer therapies into survivorship frameworks. Most current guidelines have not yet incorporated the cardiovascular risks associated with newer oncology agents such as immune checkpoint inhibitors and tyrosine kinase inhibitors, despite growing evidence of their cardiotoxic potential [50]. Similarly, the prognostic role of conventional biomarkers remains uncertain. Data from registries such as CARDIOTOX indicate that baseline NT-proBNP and cardiac troponin have limited predictive value for later cardiotoxicity, underscoring the need for dynamic and individualized surveillance models [55]. Preventive strategies also remain underemphasized. Landmark studies, including INTERHEART and PURE, demonstrated that modifiable lifestyle and metabolic factors account for most of the global cardiovascular disease burden [56-58]. This evidence is particularly relevant for cancer survivors, whose prior therapies often exacerbate hypertension, dyslipidemia, and other risk factors. Strengthening preventive cardiology within survivorship programs supported by exercise, diet, and behavioral interventions could substantially improve long-term outcomes. Therapy-specific risks continue to shape survivorship care. Historical radiotherapy regimens delivered mean heart doses (MHD) of 5–10 Gy, strongly linked to late ischemic heart disease [59-60]. Modern techniques such as three-dimensional conformal planning, intensity-modulated radiotherapy (IMRT), and deep-inspiration breath hold (DIBH) have reduced MHDs to below 2 Gy in most left-sided breast cancer patients [61-62]. Similarly, anthracycline-induced cardiotoxicity remains dose-dependent despite

mitigation through liposomal formulations, dexrazoxane, and neuro-hormonal blockade trials such as PRADA, CECCY, and ICOS-ONE, which show partial protective benefit [63-65]. Looking forward, survivorship care must advance beyond static guideline frameworks. The next phase of cardio-oncology should integrate precision risk prediction, survivor-specific biomarkers, and adaptive surveillance intervals tailored to individual risk.

Limitations

There are several limitations that need to be discussed. Although we used an extended search strategy, and consulted all websites of professional societies, we may have missed guidelines with recommendations for CVD in cancer survivors. Further, more evidence on specific interventions may have been published since the publication of the guidelines and therefore may not have been included in this study. As that newer evidence is also not included in the guidelines, clinicians are also not likely to use them for their patients. Given that the clinical focus of this review is CVD, it might ordinarily be expected that the keywords “CVD-related terms” be included in the search. However, our rationale for not doing so was that recommendations on CVD are frequently embedded within cancer-type-specific survivorship guidelines (e.g., breast, prostate, or colorectal cancer) that do not necessarily mention fatigue in their titles or abstracts. Limiting the search to CVD-specific terminology would therefore have risked excluding a substantial number of relevant guidelines. To mitigate this, we employed a deliberately broad search strategy designed to capture all survivorship and follow-up guidelines, irrespective of cancer type, and subsequently performed full-text screening to extract recommendations related to CVD. In addition, the included guidelines span more than two decades, during which both the strength of the evidence base and the methodological standards for guideline development have evolved. As a result, some variability in recommendations may be attributed to publication date. Older guidelines may not reflect current best evidence and should therefore be interpreted with caution. Importantly, our analysis was restricted to recommendations for post-treatment survivorship care. We did not evaluate interventions specific to patients undergoing active cancer treatment, nor those living with advanced disease or receiving end-of-life care. Thus, this study provides the most comprehensive overview of current recommendations and their evidence for the management of CVD in the survivorship phase.

Implications for Research

Future research must move beyond acute cardiotoxicity monitoring toward long-term cardiovascular risk prediction and prevention strategies. Despite the growing survivor population, relatively few studies address late-onset cardiovascular disease. Prospective, longitudinal cohorts are needed to define optimal surveillance intervals, evaluate cardio-protective interventions, and clarify cumulative treatment effects.

Current tools such as the Framingham [66] and ASCVD risk scores [67] underestimate risk in cancer survivors. Developing survivor-specific prediction models, ideally leveraging machine learning to integrate treatment exposures and comorbidities, could enable more tailored surveillance and therapeutic strategies. Evidence supporting biomarkers remains limited. Studies should validate molecular and imaging markers that capture vascular injury, fibrosis, or microvascular dysfunction, alongside multi-omics approaches to identify early signals of subclinical damage.

Trials of preventive pharmacotherapy, including statins, beta-blockers, and ACE inhibitors are needed to determine efficacy, dosing, and treatment duration in asymptomatic survivors. In parallel, research must define optimal lifestyle interventions, including exercise and anti-inflammatory nutrition, and assess novel pathways such as the gut–heart axis. Integrating these advances will enable cardio-oncology to transition from reactive management of cardiotoxicity to proactive, personalized cardiovascular prevention across survivorship. Finally, emerging oncologic therapies such as immunotherapies and targeted agents are increasingly used in curative settings but remain insufficiently integrated into current risk stratification models, despite their known potential to cause cardiovascular complications [68-69].

Implications for Clinical Practice

Cardiovascular health should be an integral component of survivorship care. Risk stratification at the time of treatment initiation, along with periodic reassessment during follow-up, can help identify survivors at increased risk of CVD and guide the intensity of monitoring and intervention. Oncologists should provide this information at the end of treatment to support PCPs in coordinating ongoing care. Routine cardiovascular assessments, including clinical evaluations and echocardiography, should be

considered for high-risk survivors, even in the absence of symptoms. When symptoms such as dyspnea, orthopnea, or reduced exercise tolerance emerge, timely referral to cardiology or cardio-oncology specialists is essential. Lifestyle counseling should be routinely integrated into survivorship visits. Survivors should be encouraged to engage in regular physical activity, adopt heart-healthy diets, and manage traditional risk factors such as hypertension, dyslipidemia, diabetes, and smoking. Embedding cardiovascular risk assessment and prevention into survivorship care can reduce long-term morbidity and improve overall health outcomes.

Conclusion

This analysis highlights a consensus among professional societies on the central role of risk stratification in the management of cardiovascular health in cancer survivors, serving as the cornerstone for clinical decision-making. Despite this, the translation of recommendations into routine clinical practice remains challenging. Existing guidelines vary across oncology, cardiology, and general medicine, reflecting differences in disciplinary focus, methodological approaches, and evidentiary standards. This heterogeneity may impede interdisciplinary collaboration. Moreover, numerous guidelines have yet to fully integrate data on novel oncologic therapies or the role of structured rehabilitation programs, underscoring the rapidly evolving nature of the evidence base. Consequently, harmonized, consistently updated, and robust, evidence-based interdisciplinary recommendations are essential to improve cardiovascular outcomes in cancer survivors.

Additionally, there is an urgent need for further research into survivor-specific cardiovascular risk prediction, the development and validation of novel biomarkers, and the integration of multi-omics approaches to support personalized survivorship care.

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Author Contribution

J.A., K.G-F., E.H-L., and M.M.W. conceived the study and designed the methodology. J.A. and N.S. conducted the literature search, data extraction, analysis, and quality assessments. E.H-L. and S.I.R. provided guidelines from external websites, S.I.R. provided oncological expertise and a critical review of clinical relevance. J.A. wrote the main manuscript text. E.H-L. and M.M.W. reviewed the manuscript. J.A., N.S., N.M., B.C., R.T, L.K., L.S., S.I.R., K.G-F., E.H-L., M.M.W. contributed to the final manuscript revisions and approved the final version for submission.

Conflict of Interest Statement

J.A declare no competing interest. N.S declare no competing interest. N.M declare no competing interest. B.C declare no competing interest. L.K declare no competing interest. L.S declare no competing interest. R.T declare no competing interest. S.R declare competing interest; Research support from other organizations: Roche, AstraZeneca, Swiss National Foundation and Amgen. K.G-F declare competing interest; Grant from the Kanton Aargau, Switzerland. Funding a pilot project for consultation of cancer survivors. E.H-L declare competing interest; Grant from the Kanton Aargau, Switzerland. Funding a pilot project for consultation of cancer survivors. M.M.W declare competing interest; Grant from the Kanton Aargau, Switzerland. Funding a pilot project for consultation of cancer survivors and a Grant from the Stiftung Cardio.”

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Figure 1. Preferred reporting items for systematic reviews and meta-analysis statement flow chart.

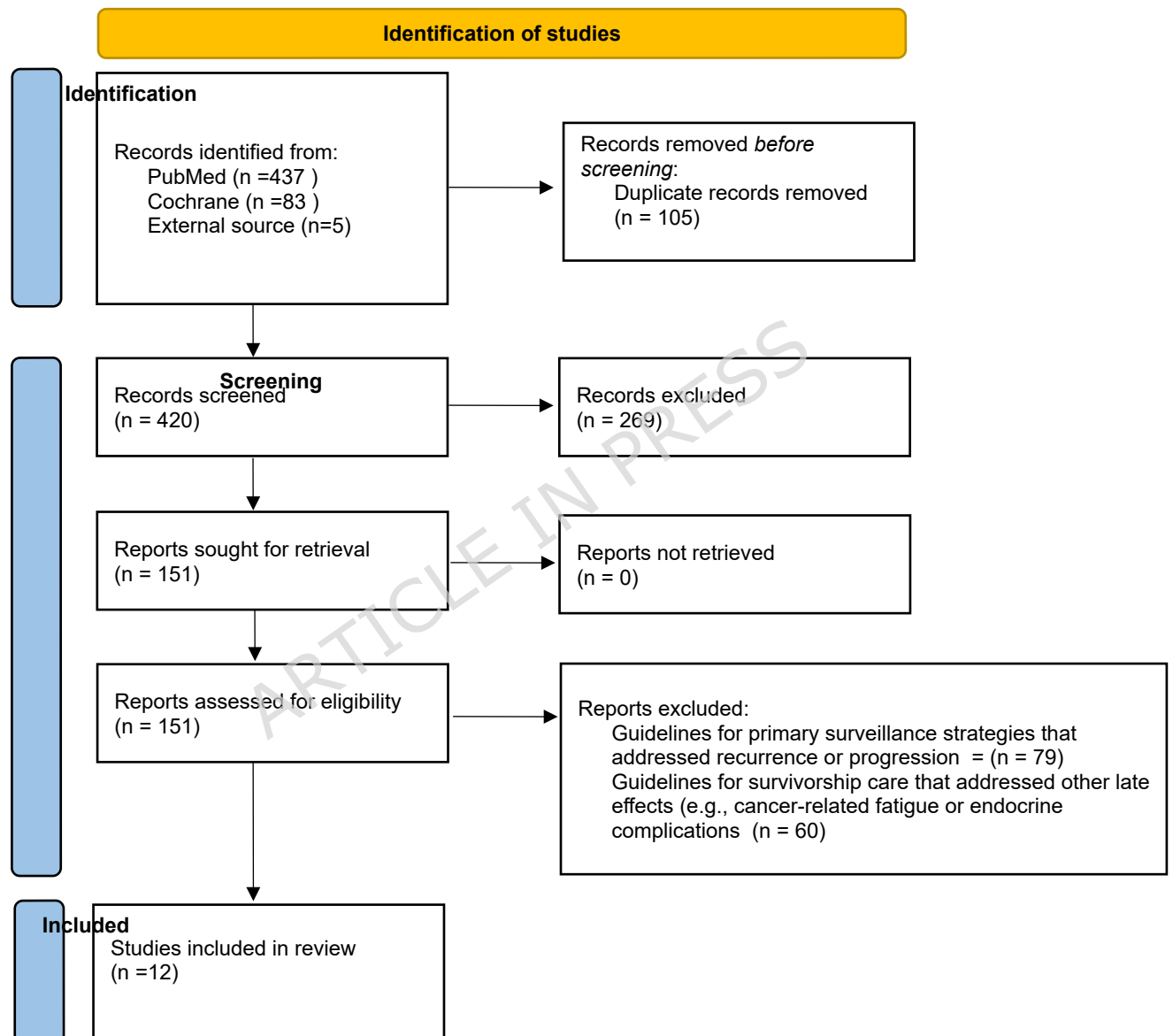


Table 1. Description of Guidelines on CVD in Survivorship Care

Guideline	Region	Issuing Organization	Scope	Cancer Type-Specific	Overall quality
ASCO/ACS, 2016 [34]	USA	American Cancer Society (ACS)	Cancer type-specific	Breast Cancer	Moderate
ASCO, 2015 [36]	USA	American Society of Clinical Oncology (ASCO)	Cancer type-specific	Colorectal Cancer	Moderate
ACS, 2014 [35]	USA	American Cancer Society (ACS)	Cancer type-specific	Prostate cancer survivors	Moderate
ACSM, 2019 [20]	USA	American College of Sports Medicine (ACSM)	Broad	Broad	Moderate
ESMO, 2022 [37]	Europe	European Society of Medical Oncology (ESMO)	Broad	Broad	Moderate
ESMO, 2012 [38]	Europe	European Society of Medical Oncology (ESMO)	Broad	Broad	Moderate
ESMO, 2024 [39]	Europe	European Society of Medical Oncology (ESMO)	Cancer type-specific	Breast Cancer	Moderate
ASCO, 2017 [16]	USA	American Cancer Society (ACS)	Broad	Broad	High
ESC, EHA, ESTRO, IC-OS 2022 [18]	Europe	EHA, European Hematology Association. ESC, European Society of Cardiology. ESTRO, European Society for Therapeutic Radiology and Oncology. IC-OS, International Cardio-Oncology Society	Broad	Broad	High
NCCN, 2024 [17]	USA	National Comprehensive Cancer Network (NCCN)	Broad	Broad	Moderate
SEOM 2018 [19]	Europe	Spanish Society of Medical Oncology.	Cancer type-specific	Breast Cancer	Moderate
ESMO 2020 [40]	Europe	European Society of Medical Oncology (ESMO)	Broad	Broad	Moderate

Table 2. Guideline recommendation for the screening for cardiovascular diseases and clinical assessment.

General Risk Stratification	Guidelines-year	Level of evidence	Strength of recommendations
A risk stratification of asymptomatic cancer survivors is recommended based on their baseline CV risk, type and cumulative dose of cancer therapy (particularly anthracyclines and radiation), timing since treatment, and presence or recovery of cardiac dysfunction. The guideline recommended the use of cardiovascular risk estimation tools such as SCORE2 and SCORE2-OP. Risk categories are: Very high risk, early high risk, late high risk, moderate risk, and low risk.	ESC 2022[18]	Consensus based	NS
Encourage the use of a validated risk stratification tool (e.g. the ASCVD risk tool) to assess the 10-year cardiovascular risk and the lifetime risk for cardiovascular disease	NCCN 2024[17]	Moderate	NS
A risk stratification of survivors is recommended based on treatment type and dose, baseline heart risk, additional risk factors, and combined or sequential therapies.	ASCO 2017[16]	Moderate	Conditional for
A risk stratification of survivors is recommended based on cumulative doses of anthracyclines.	ESMO 2012[38]	High	Strong for
Recommend the evaluation of modifiable risk factors (e.g., obesity, hypertension, dyslipidemia).	NCCN 2024[17]	Moderate	NS
For men receiving androgen deprivation therapy, it is recommended to screen for cardiovascular risk factors following the United States Preventive Services Task Force guidelines for: - Hypertension - Type 2 diabetes - Lipid disorders	ACS 2014[35]	Strong Moderate Strong	NS
<i>Specific risk stratification based on cardiotoxicity of certain treatment</i>			

Risk stratification	ESC 2022 [18]		
Very High Risk: Very high baseline CVD risk before treatment, cumulative doxorubicin doses ≥ 400 mg/m², radiotherapy (RT) with mean heart dose (MHD) > 25 Gy, or combined RT 15–25 Gy with doxorubicin ≥ 100 mg/m².	ESC 2022 [18]	Consensus based	NS
Early High Risk: High baseline CV risk, defined as those who developed moderate-to-severe CTRCD during treatment (symptomatic or not), doxorubicin 250–399 mg/m², or high-risk hematopoietic stem cell transplant (HSCT) recipients.			
Late High Risk: Includes patients with moderate past exposure (e.g., RT 15–25 Gy or RT 5–15 Gy plus doxorubicin ≥ 100 mg/m²) and/or poorly controlled CV risk factors.			
Moderate Risk: Includes those with moderate baseline CV risk, doxorubicin 100–249 mg/m², RT 5–15 Gy, or RT < 5 Gy combined with doxorubicin ≥ 100 mg/m².			
Low Risk: Encompasses patients with low baseline CV risk, normal post-therapy cardiac assessment, doxorubicin < 100 mg/m², RT < 5 Gy, and those who had only mild, fully resolved CTRCD during treatment. The ESC guideline further recommends re-stratifying CVD risk five years after cancer treatment, especially in survivors of thoracic radiation or high-dose anthracyclines, recognizing the progressive nature of late radiation-induced cardiovascular injury.			
There is an increased risk for developing cardiac dysfunction in survivors with treatment that included any of the following: - High-dose anthracycline (e.g, doxorubicin ≥ 250 mg/m², epirubicin ≥ 600 mg/m²) - High-dose RT (≥ 30 Gy) where the heart is in the treatment field - Lower-dose anthracycline (e.g, doxorubicin 250 mg/m², epirubicin 600 mg/m²) in combination	ASCO 2017 [16]	Moderate	Conditional for

with lower dose RT (30 Gy) where the heart is in the treatment field There is also an increased risk if patient was treated with lower-dose anthracycline (e.g., doxorubicin 250mg/m², epirubicin 600mg/m²) or trastuzumab alone, and presence of any of the following risk factors: • Multiple cardiovascular risk factors ($\geq$ two risk factors), including smoking, hypertension, diabetes, dyslipidemia, and obesity, during or after completion of therapy • Older age ($\geq$ 60 years) at cancer treatment • Compromised cardiac function (e.g, borderline low LVEF [50% to 55%], history of myocardial infarction, $\geq$ moderate valvular heart disease) at any time before or during treatment • Treatment with lower-dose anthracycline (e.g, doxorubicin 250 mg/m², epirubicin 600 mg/m²) followed by trastuzumab (sequential therapy)			
Patients should be considered at risk for cardiac toxicity in case they have history of exposure to any of the following cumulative doses of anthracyclines as specified below. • Doxorubicin >500 mg/m² • Liposomal doxorubicin >900 mg/m² • Epirubicin >720 mg/m² • Mitoxantrone >120 mg/m² • Idarubicin >90 mg/m²	ESMO 2012 [38]	High	Strong for
Surveillance			
Annual CV risk assessment (clinical review, BP, lipid profile, HbA1c), including ECG and NP, and CVRF management is recommended in CS who were treated with a potentially cardiotoxic cancer drug or radiation therapy.	ESC 2022[18]	Moderate	Strong for
Risk re-stratification is recommended 5 years after therapy to organize long-term follow-up.	ESC 2022[18]	Low	Strong for

In asymptomatic patients with normal cardiac function who have received potentially cardiotoxic treatment, cardiac follow-up should be carried out as clinically indicated	ESMO 2024[39]	low	Conditional for
It is recommended that primary care clinicians should monitor lipid levels and provide cardiovascular monitoring, as indicated in breast cancer survivors	ASCO/ACS 2016[34]	Very low	NS
Recommendations for echocardiography depending on the risk stratification:	ESC 2022[18]		
• very high and early high-risk: At year 1, 3, and 5 after completion of cardiotoxic cancer therapy and every 5 years thereafter should be considered in asymptomatic	ESC 2022[18]	Low	Conditional for
• Asymptomatic late high-risk: Should be considered starting at 5 years after radiation to a volume including the heart and then every 5 years.	ESC 2022[18]	Low	Conditional for
• Asymptomatic moderate-risk: Echocardiography may be considered every 5 years	ESC 2022[18]	Low	Conditional for
Survivors from node-positive, human epidermal growth factor receptor 2-positive breast cancer with high dose exposure or additional risk factors: Consider echocardiography within 1-year post-anthracycline therapy	NCCN 2024[17]	Moderate	NS
For coronary artery disease (CAD)	ESC 2022[18]		
- Screening for CAD should be considered in high-risk patients who have received chest RT to a treatment volume including the heart. - Non-invasive testing should be considered every 5–10 years in asymptomatic patients who received >15 Gy MHD starting at 5 years after radiation. - Screening for CAD should be considered in high-risk patients who have received chest RT to a treatment volume including the heart. - Screening for CAD includes stress echocardiography, cardiac CT, stress CMR, single-photon emission CT stress test, according to local protocol.	ESC 2022[18]	Low	Conditional for
For patients with a history of mediastinal chest RT, evaluation for CAD and ischaemia, as well as valvular disease is recommended, even if asymptomatic, starting at 5 years post-treatment and then at	ESMO 2020[40]	High	Strong for

least every 3-5 years thereafter			
In individuals with clinical signs or symptoms concerning cardiac dysfunction during routine clinical assessment, test for serum cardiac biomarkers (troponins, natriuretic peptides) or echocardiography-derived strain imaging in conjunction with routine diagnostic imaging should be recommended	ASCO 2017[16]	Moderate	Conditional for
Patients treated for metastatic disease: LVEF should be monitored at baseline and then infrequently in the absence of symptoms	ESMO 2012[38]	Moderate	Strong for
Monitor colorectal cancer survivors who are obese or who have had prior coronary artery disease and received 5-fluorouracil or capecitabine for cardiovascular disease.	ACS 2015[36]	Very low	NS
Carotid ultrasound imaging should be considered every 5 years in asymptomatic patients with a history of head/neck RT, starting at 5 years after radiation and every 5–10 years thereafter.	ESC 2022 [18]	Low	Conditional for
For asymptomatic patients who have been treated with cardiotoxic agents and have normal cardiac function, periodic screening for the development of new asymptomatic left ventricular dysfunction with cardiac biomarkers and potentially cardiac imaging should be considered at 6 to 12 months, at 2 years post-treatment and possibly periodically thereafter	ESMO 2020[40]	Moderate	Strong for
For pregnancy	ESC 2022[18]		
In high-risk female CS, pre-pregnancy counseling and management during pregnancy and around delivery by a multidisciplinary pregnancy heart team is recommended.	ESC 2022[18]	Low	Strong for
A baseline CV evaluation including history, physical examination, ECG, NP, and echocardiography is recommended in female CS with a history of CTRCD who are considering pregnancy.	ESC 2022[18]	Low	Strong for
A baseline CV evaluation including history, physical examination, ECG, and echocardiography should be considered in all female CS who received potentially cardiotoxic cancer therapy and are considering pregnancy.	ESC 2022[18]	Low	Conditional for
A CV evaluation including echocardiography is recommended at 12 weeks of pregnancy in female CS who are either high-risk or who received potentially cardiotoxic cancer therapy and did not have a baseline CV assessment.	ESC 2022[18]	Low	Strong for

A second CV evaluation including echocardiography should be considered at 20 weeks of pregnancy in high-risk female CS who received potentially cardiotoxic cancer therapy.	ESC 2022[18]	Low	Conditional for
Assessment			
Clinicians should complete a careful history and physical examination in survivors of cancer previously treated with potentially cardiotoxic therapies	ASCO 2017[16]	Very low	Strong for
Assess symptoms like orthopnea, paroxysmal nocturnal dyspnea, and activity tolerance in survivors. A history and physical evaluation should be performed to look for these symptoms to help identify survivors who might already be symptomatic.	NCCN 2024[17]	Moderate	NS
Evaluate for signs of volume overload and structural heart disease in symptomatic individuals.	NCCN 2024[17]	Moderate	NS
In individuals with clinical signs or symptoms for cardiac dysfunction during routine clinical assessment, the following strategy is recommended:	ASCO 2017 [16]		
- TTE should be recommended for diagnostic workup	ASCO 2017 [16]	Moderate	Strong for
- If TTE is not available or technically not feasible (e.g., poor quality), use cardiac magnetic resonance imaging (MRI, preferred) or multigated acquisition (MUGA) scan	ASCO 2017 [16]	Moderate	Conditional for
In symptomatic survivors, TTE should be performed.	NCCN 2024[17]	Moderate	NS
- Symptomatic survivors with evidence of structural heart disease require immediate referral to a cardio-oncologist or cardiologist.			
- If no evidence of structural heart disease is seen, workup for other causes of the symptoms is warranted with referral to other specialties (e.g., pulmonology or cardiology) as needed.			
Care coordination			
Referral to a cardiologist based on findings. Patients identified to have asymptomatic cardiac dysfunction during routine surveillance should be referred to a cardiologist or a health care provider with cardio-oncology expertise for further assessment and management.	ASCO 2017 [16]	Very low	Strong for

Refer to cardiology if signs of heart failure appear	SEOM 2018[18]	NS	NS
Identify and refer survivors with symptoms or evidence of cardiomyopathy (Stage B, C, or D) for specialist care. Refer survivors with high cardiovascular risk or symptoms to a cardiologist or cardio-oncologist if available.	NCCN 2024[17]	Moderate	NS

Legend: ACS, American Cancer Society; ASCO, the American Society of Clinical Oncology; ASCVD, atherosclerotic cardiovascular disease; CAD, coronary artery disease; CCTA, cardiac computed tomography angiography; CMR, cardiac magnetic resonance imaging; CTR-CVT, cancer therapy-related cardiovascular toxicity; NS, not stated; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; ESC, European Society of Cardiology; ESMO, European Society for Medical Oncology; Gy, Gray (unit of absorbed radiation dose); HbA1c, glycated hemoglobin; HSCT, haematopoietic stem cell transplant; LVEF, left ventricular ejection fraction; MHD, mean heart dose; MRI, magnetic resonance imaging; MUGA, multigated acquisition scan; NCCN, National Comprehensive Cancer Network; NP, natriuretic peptides; NS, not specified; RT, radiotherapy; SEOM, Sociedad Española de Oncología Médica.

Table 3. Guideline recommendation for the Management of Specific Cardiovascular Conditions in Cancer Survivors

	Guideline	Level of evidence	Strength of recommendations
Lifestyle modification			
Promote physical activity, healthy diet, smoking cessation, and weight control as key survivorship strategies.	NCCN 2024[17]	Moderate	NS
Exercise to reduce cardiotoxicity (no details)	ACSM 2019[20]	Low	NS
Promotion of healthy lifestyles	SEOM 2018[18]	NS	NS
Clinicians should regularly evaluate and manage cardiovascular risk factors such as smoking, hypertension, diabetes, dyslipidemia, and obesity in patients previously treated with cardiotoxic cancer therapies. A heart-healthy lifestyle, including the role of diet and exercise, should be discussed as part of long-term follow-up care.	ASCO 2017[16]	Moderate	Conditional for
It is recommended that primary care clinicians should educate breast cancer survivors on healthy lifestyle modifications, potential cardiac risk factors, and when to report relevant symptoms (shortness of breath or fatigue) to their health care provider	ASCO/ACS 2016[34]	High	NS
Patients undergoing anticancer therapy and long-term cancer survivors should be encouraged to exercise on a regular basis	ESMO 2020[40]	Moderate	Conditional for
Patients undergoing anticancer therapy and long-term cancer survivors should be encouraged to have healthy dietary habits (high intake of fresh fruits/vegetables and whole grains as compared with refined grains, processed and red meats and high-fat foods) and to maintain a normal weight	ESMO 2020[40]	Low	Strong for
General recommendation			
Address comorbidities such as hypertension, diabetes, and dyslipidemia as part of CVD prevention.	NCCN 2024[17]	Moderate	NS
Control concomitant diseases (hypertension, diabetes, obesity)	SEOM 2018[18]	NS	NS
In men receiving ADT, use the USPSTF guidelines to address and educate about cardiovascular risk		Strong for	NS

factors, and treat high blood pressure, dyslipidemia, and diabetes.	ACS 2014[35]	hypertension, Moderate for type II diabetes, Strong for lipid disorders	
Asymptomatic radiation-induced CAD			
Non-invasive stress testing is recommended in asymptomatic CS with new moderate or severe radiation-induced CAD detected on CCTA to guide ischemia-directed management.	ESC 2022[18]	Very low	Strong for
A multidisciplinary team (MDT) discussion is recommended for clinical decision-making in patients with radiation-induced CAD and inducible ischemia or severe left main CAD.	ESC 2022[18]	Insufficient	Strong for
Symptomatic CAD			
In patients with radiation-induced CAD where CABG surgery is considered, pre-operative assessment of LIMA and RIMA viability, venous access, and sternal wound healing is recommended.	ESC 2022[18]	Very low	Strong for
PCI may be considered in CS with radiation-induced CAD with severe left main or three-vessel disease with a high SYNTAX score (>22) in whom the procedure is technically feasible.	ESC 2022[18]	Low	Conditional for
Other causes of troponin elevation should be ruled out, including ACS if appropriate (patients with CV risk factors or established coronary artery disease)	ESMO 2022[37]	Very low	Strong for
Valvular heart disease			
A multidisciplinary team including cardiac surgeons, interventional cardiologists, and cardio-oncology specialists' approach is recommended to discuss and define the surgical risk in CS with severe VHD.	ESC 2022[18]	Very low	Strong for
TAVI should be considered for patients with symptomatic severe aortic stenosis caused by radiation at intermediate surgical risk.	ESC 2022[18]	low	Conditional for
Cancer therapy-related cardiac dysfunction			
ACE-I/ARB and/or beta-blockers are recommended in adult CS with moderate asymptomatic cancer therapy-related cardiac dysfunction	ESC 2022[18]	high	Strong for

ACE-I/ARB and/or beta-blockers may be considered in adult CS with mild asymptomatic cancer therapy-related cardiac dysfunction.	ESC 2022[18]	low	Conditional for
For patients who developed LVD or HF due to trastuzumab (or any HER2-targeted molecular therapy), anthracyclines or other anticancer therapies, CV care including medical treatment with ACE-Is, ARBs and/or BBs and regular cardiology review (e.g. annual if asymptomatic) should be continued indefinitely, regardless of improvement in LVEF or symptoms. Any decision to withdraw HF-based therapy should only be done after a period of stability, no active cardiac risk factors and no further active anticancer therapy	ESMO 2020[40]	Moderate	Strong for

Legend: ACS, American Cancer Society; ACE-I, angiotensin-converting enzyme inhibitor; ACSM, American College of Sports Medicine; AD, androgen deprivation therapy; ADT, androgen deprivation therapy; ARB, angiotensin II receptor blocker; ASCO, American Society of Clinical Oncology; ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CABG, coronary artery bypass grafting; CAD, coronary artery disease; CCTA, cardiac computed tomography angiography; CMR, cardiac magnetic resonance imaging; CS, cancer survivors; CTR-CVT, cancer therapy-related cardiovascular toxicity; CTRCD, cancer therapy-related cardiac dysfunction; CV, cardiovascular; CVD, cardiovascular disease; CVRF, cardiovascular risk factors; ECG, electrocardiogram; ESC, European Society of Cardiology; ESMO, European Society for Medical Oncology; Gy, Gray (unit of radiation dose); HbA1c, glycated hemoglobin; HSCT, haematopoietic stem cell transplant; LIMA, left internal mammary artery; LVEF, left ventricular ejection fraction; MDT, multidisciplinary team; MHD, mean heart dose; MRI, magnetic resonance imaging; MUGA, multigated acquisition scan; NCCN, National Comprehensive Cancer Network; NP, natriuretic peptides; NS, not stated; PCI, percutaneous coronary intervention; RIMA, right internal mammary artery; RT, radiotherapy; SYNTAX, Synergy Between PCI With Taxus and Cardiac Surgery score; TAVI, transcatheter aortic valve implantation; USPSTF, United States Preventive Services Task Force; VHD, valvular heart disease.

Supplementary table

Table S1. - Search complete 16.01.2025.

Table S1	Search strategy - 28.12.2024	
PUBMED		
Step	Search strategy	References
1	("cancer survivors"[MeSH Terms] OR "survivorship"[All Fields] OR "post-cancer care"[All Fields] OR "cancer"[All Fields] OR "follow-up"[All Fields] OR "after therapy"[All Fields]) AND ((guideline[Filter]) AND (2000:2024[pdat]))	6510
2	((("guidelines as topic"[MeSH Terms] OR "clinical practice guidelines"[All Fields]) AND "follow-up"[All Fields]) OR "survivorship"[All Fields]) AND ((guideline[Filter]) AND (2000:2024[pdat]))	468
3	("childhood"[All Fields] OR "pediatric"[All Fields] OR "young adult"[All Fields]) AND "cancer"[All Fields]) AND ((guideline[Filter]) AND (2000:2024[pdat]))	378
4	#1 AND #2 NOT #3	437
COCHRANE		
1	("cancer survivorship" or "post-cancer care" or "cancer follow-up" or "after cancer treatment")	1403
2	("clinical practice guideline" or "recommendation" or "consensus" or "clinical practice guidelines")	29794
3	("childhood" or "pediatric" or "young adult")	199464
4	#1 AND #2 NOT #3. 2000-2024	83

Table S2. Quality analysis assessment of the guidelines with domains according to AGREE II.

Society, Author (year)	Coverage	1	2	3	4	5	6	OQ
ESMO 2012[38]	Survivors with immune related toxicity	**	**	*	**	***	**	**
ESMO 2022[37]	All cancer survivors	**	**	**	***	**	**	**
ACS/ASCO 2016[34]	Survivors of breast cancer	***	**	**	***	*	***	**
ASCO 2014[35]	Survivors of prostate cancer	***	**	***	**	*	***	**
NCCN, 2024[17]	All cancer survivors	***	**	**	***	**	**	**
ESMO 2024[39]	Survivors of breast cancer	**	**	**	***	**	***	**
ACS 2017[16]	All cancer survivors	***	***	***	***	**	**	***
ESC, EHA, ESTRO, IC-OS 2022 [18]	All cancer survivors	***	**	***	***	**	***	***
ACSM 2019[20]	All cancer survivors	***	**	**	***	*	**	**
SEOM 2018[19]	Survivors of breast cancer	***	**	**	***	**	**	**
ACS 2015[36]	Survivors of colorectal cancer	***	**	***	**	*	***	**
ESMO 2020[40]	All cancer survivors	***	**	**	***	*	***	**

NOTE. Quality of a domain was defined as good if the score was >80%-100%, moderate if the score was between 50%-80%, and poor if the score was <50% (***, good; **, moderate; *, poor). 1, Scope and Purpose; 2, Stakeholder Involvement; 3, Rigor of Development; 4, Clarity of Presentation; 5, Applicability; 6, Editorial Independence. Abbreviations: AGREE, Appraisal of Guidelines for Research and Evaluation; OQ, overall quality. Abbreviations: ESMO, European Society of Medical Oncologist. ACS/ASCO, American Cancer Society/American Society of Clinical Oncology. NCCN, National Comprehensive Cancer Network. EHA, European Hematology Association. ESC, European Society of Cardiology. ESTRO, European Society for Therapeutic Radiology and Oncology. IC-OS, International Cardio-Oncology Society. ACSM, American College of Sports Medicine. SSMO, Spanish Society of Medical Oncology.

Table S3. Definition of strength of recommendation according to the GRADE system

Predefined grading system	Strong for	Conditional (for)	Strong (against)	Conditional (against)
ESMO 2022[37]	A	B-C	D	E
ESMO 2012[38]	A	B	B	C
ACS/ASCO 2016[34]	N.S	N.S	N.S	N.S
ASCO 2014[35]	N.S	N.S	N.S	N.S
NCCN, 2024[18]	N.S	N.S	N.S	N.S
ESMO 2024[39]	I	II	Ila-IIb	III
ASCO 2017[16]	High	Intermediate	Low	Insufficient
ESC 2022 [18]	I	II, Ila, IIb	III	N.S
ACSM 2019 [20]	N.S	N.S	N.S	N.S
SEOM 2018[19]	N.S	N.S	N.S	N.S
ACS 2015[34]	N.S	N.S	N.S	N.S
ESMO 2020[40]	A,B	C	D	E
N.S, not stated				

Table S4. Definition of strength of evidence according to the GRADE system

Predefined grading system	High	Moderate	Low	Very low
ESMO 2022[37]	I	II	III-IV	V
ESMO 2012[38]	I	II	III-IV	V
ACS/ASCO 2016[34]	I, IA, IB, IC	IIA, IIB, IIC	III	0
ASCO 2014[35]	I, IA, IB, IC	IIA, IIB, IIC	III	0
NCCN, 2024[18]	I	2A	N.S	N.S
ESMO 2024[39]	A	B	B	C
ASCO 2017[16]	Strong	Moderate	Insufficient	N.S
ESC 2022 [18]	A	B	C	N.S
ACSM 2019 [20]	Strong	Moderate	Insufficient	N.S
SEOM 2018[19]	N.S	N.S	N.S	N.S
ACS 2015[34]	I, IA, IB, IC	IIA, IIB, IIC, 2A	III	0
ESMO 2020[40]	I	II,III	IV	V
N.S, not stated				

Table S5. Abbreviations

ACS	American Cancer Society
ACE-I	Angiotensin-Converting Enzyme Inhibitor
ACSM	American College of Sports Medicine
ADL	Activities of Daily Living
AGREE II	Appraisal of Guidelines for Research & Evaluation II
ARB	Angiotensin II Receptor Blocker
ARIC	Atherosclerosis Risk in Communities
ASCO	American Society of Clinical Oncology
ASCVD	Atherosclerotic Cardiovascular Disease
BB	Beta-blocker
BNP	B-type Natriuretic Peptide
CABG	Coronary Artery Bypass Grafting
CAD	Coronary Artery Disease
CMR	Cardiac Magnetic Resonance Imaging
CS	Cancer Survivors
CTRCD	Cancer Therapy–Related Cardiac Dysfunction
CVD	Cardiovascular Disease
CVRF	Cardiovascular Risk Factor
DLVD	Dysfunction of the Left Ventricle

ECG	Electrocardiogram
ECHO	Echocardiogram
ESC	European Society of Cardiology
ESMO	European Society of Medical Oncologist
GRADE	Grading of Recommendations Assessment, Development, and Evaluation
HF	Heart Failure
HER2	Human Epidermal Growth Factor Receptor 2
ICI	Immune Checkpoint Inhibitor
IR-myocarditis	Immune-Related Myocarditis
LVEF	Left Ventricular Ejection Fraction
LVH	Left Ventricular Hypertrophy
MHD	Mean Heart Dose
MDT	Multidisciplinary Team
MRI	Magnetic Resonance Imaging
MUGA	Multi-Gated Acquisition Scan

NCCN	National Comprehensive Cancer Network
NS	Not Stated
PCP	Primary Care Provider
PND	Paroxysmal Nocturnal Dyspnea
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RCT	Randomized Controlled Trial
Radiation Therapy	RT
SCP	Survivorship Care Plan
SEOM	Spanish Society of Medical Oncology
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology
TAVI	Transcatheter Aortic Valve Implantation
VHD	Valvular Heart Disease